tcpdump and Wireshark compared

Both tools capture and read network traffic. I used tcpdump for the DNS and ICMP report and Wireshark for the SYN flood log, so this is my own comparison after using each one.

Aspect	tcpdump	Wireshark
Interface	Command line (CLI)	Graphical interface (GUI)
Resources	Lightweight, runs fine on a server with no desktop	Needs more memory and CPU
Features	Basic filters and options, plain text output	Color coding, advanced filtering, following a conversation, statistics
Learning curve	I have to remember the options and the filter syntax	Easier to explore by clicking around

What they have in common

Both are network protocol analyzers. Both are open source and free. Both can filter for a specific protocol and both can save a capture to a file and read it again later.

A few commands I used

tcpdump, capture only DNS traffic on one interface and show numbers instead of names:

```
tcpdump -i eth0 -nn udp port 53
```

tcpdump, save a capture to a file, then read it back:

```
tcpdump -i eth0 -w capture.pcap
```

```
tcpdump -r capture.pcap
```

Wireshark display filter, show only DNS packets:

```
dns
```

Wireshark display filter, show the first packet of each TCP handshake (useful to spot a SYN flood):

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

When I would use which

I would use tcpdump when I am on a server over a remote connection, or when I just need a quick capture saved to a file. I would open that file in Wireshark when I need to follow one conversation or look at a lot of packets at once, because reading them there is much less tiring than in a terminal.